USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 6 LAB - AM Simulation
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.1: I can describe DSB-AM, DSB-SC, and SSB and identify each on a spectrum.
Learning Objective 3.2: I can implement an AM modulator and demodulator in GNU Radio and verify demodulated audio against the source.

This packet is the L6 Lab, in three parts, and you turn it in once, complete, before Lesson 7. Do **Part 1** on your own before the lab period: predict every spectral line and receiver effect you will measure. **Part 2** is your own measurements, on your own laptop, recorded in class. Do **Part 3** on your own after the lab: compare the measurements with the predictions and answer the analysis questions.

Grading. One submission, graded against Objectives 3.1 and 3.2 on the course's 1 / 0 scale, with revision allowed under course policy. Paste screenshots where a question asks for one.

No hardware. Every flowgraph in this lab is a simulation. Nothing transmits, and you need only your laptop and its speakers or headphones. Start with the volume low.

The flowgraphs are on the L6 lesson page: `L06-dsb-am.grc`, `L06-dsb-sc.grc`, and `L06-ssb.grc`. All three use the same message, $x(t) = 0.6\cos(2\pi \cdot 1000\,t) + 0.4\cos(2\pi \cdot 2500\,t)$, on a carrier at $f_c = 10$ kHz, sampled at 48 kHz.

Part 1: Pre-Lab (individual, before the lab period)

1. **Predictions.** You will check each of these at the bench.

   (a) For DSB-AM with $m = 0.5$ and a carrier of amplitude 1, predict each line's frequency and its level relative to the carrier.

Line (kHz)					
Level (dB)					

   (b) List the lines you expect for DSB-SC, USB, and LSB.

(c) Find the smallest m at which this message overmodulates.

$m =$ ____

(d) Predict the DSB-SC coherent receiver's output amplitude, relative to its value with a matched LO, at each LO phase.

LO phase	0°	45°	60°	90°
Predicted				

(e) The USB receiver's LO sits 200 Hz above the carrier. Predict the two output tones.

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Part 2: Measurements (your own laptop, in class)

2. **Build the DSB-AM flowgraph.** Start GRC with an empty canvas. Do not open `L06-dsb-am.grc`: build the transmitter and the envelope detector yourself, with a `samp_rate` of 48000, an `fc` of 10000, a QT GUI Range named `m` from 0 to 1.5, a frequency sink, a time sink comparing the detector output with the delayed envelope, and an Audio Sink.

   (a) Paste a screenshot of your flowgraph on the GRC canvas. Every connection must be black, not red.

   (b) Run it at $m = 0.5$. Read each line's level off the frequency sink and subtract the carrier's level.

Line (kHz)	7.5	9	10	11	12.5
Measured (dB)					

   (c) Paste a screenshot of the time sink that compares the detector output with the delayed envelope at $m = 0.5$, and state whether they match.

3. Overmodulation. Keep your DSB-AM flowgraph running with the audio on.

(a) Raise m in steps of 0.05 and record the first value at which the detector output visibly departs from the envelope.

$m =$ ____

(b) Paste a screenshot at $m = 1.4$ and describe how the audio changes between $m = 0.5$ and $m = 1.4$.

4. DSB-SC. Open and run `L06-dsb-sc.grc`.

(a) Confirm the lines you predicted in Question 1(b), and state the level of the frequency sink at 10 kHz.

(b) Set the LO phase to each value and read the coherent receiver's peak output off the time sink.

LO phase	0°	45°	60°	90°
Peak output				

(c) Switch Listen To from Coherent to Envelope, and describe the difference.

5. SSB. Open and run `L06-ssb.grc`.

(a) For each setting of the Sideband menu, record which lines remain and how far below them the suppressed sideband sits.

Sideband	Lines (kHz)	Suppression (dB)
USB		
LSB		

(b) With USB selected, set the tuning error to $+200$ Hz. Paste the time sink and describe the audio.

(c) Find the largest tuning error, either sign, at which you cannot hear any pitch change from the original. This is the threshold of a noticeable shift, not of intelligibility.

Part 3: Writeup (individual, after the lab)

6. Analysis.

(a) Compare your measured DSB-AM levels in Question 2(b) with the predictions in Question 1(a), and explain any difference larger than 1 dB.

(b) Explain in words why an envelope detector recovers DSB-AM but not DSB-SC, and what the coherent receiver has that the envelope detector does not.

(c) A DSB-SC LO that is off in phase loses volume, while an SSB receiver that is off in frequency changes pitch. Using your measurements, explain which error a voice listener tolerates better, and why SSB receivers have a tuning knob instead of carrier recovery.

Documentation: _____